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IVS Project – (INTERCALCULATOR)

VERSION 1.0.1
Programmer Manual

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1 Current Appearance of the Calculator



2 Introduction

This document serves as a supplementary manual for developers contributing to the INTERCALCULATOR project. Reading the entire manual is strongly recommended. This document is not intended for end users of the application.

2.0.1 Getting Started

1) It is recommended to clone the project into a chosen target directory:

```
$ git clone https://github.com/honzakralem/ivs-project-2
```

2) Create a virtual environment using the make command:

```
$ make
```

3) Activate the virtual environment. This will differ depending on the operating system. For Linux/macOS:

```
$ source env/bin/activate
```

For Windows:

```
$ env\Scripts\activate
```

4) It is strongly recommended to configure your git name and email (although it is not mandatory):

```
$ git config --global user.name "name"
$ git config --global user.email "email"
```

3 Prerequisites

For the current phase of development, the following tools/programs must be installed:

- **PYTHON3**: required for running the code itself
- **GNU Make**: required for running automation scripts
- **GIT**: required for version control and collaboration with other developers via the remote repository
- **DOXYGEN**: used for generating documentation
- **GITHUB CLI**: used to download the latest installers from Releases for make pack

4 Architecture

This section covers the logic and structure of the project. Files and directories along with their roles are described here.

- **src** - source code
 - **mathlib.py** - implementation of mathematical functions

- `mathlib_test.py` - unit tests that verify the correct functionality of `mathlib.py`
 - `requirements.txt` - list of external libraries required by the project to run or for testing
 - `infixtopost.py` - converts a string from infix notation (e.g. $2 + 2$) to postfix notation (e.g. `2 2 +`)
 - `infixtopost_test.py` - unit tests for the functionality of `infixtopost.py` functions
 - `gui.py` - main graphical user interface
 - `setup.py` - script used to install dependencies
 - `Doxyfile` - configuration file for *Doxygen* documentation
 - The directory also contains `makefile` and `.gitignore`
- `mockup` - directory containing the future user interface design of the calculator
 - `plan` - directory containing the project plan and deadlines
 - `profiling` - directory containing programs for measuring calculator performance
 - The root project directory will contain `makefile`, `LICENSE`, `README.MD`, and `.gitignore`

5 Testing

This section covers testing the functionality of modules. To run the tests, simply execute the following command from any directory containing a `makefile`:

```
$ make test
```

This will run `infixtopost_test.py` and `mathlib_test.py`. The result will be reported with a message.

6 Makefile and Automation

This section deals primarily with the `makefile` in the root directory of the project. It contains commands to automate development. All commands can be run (assuming the current working directory is the project root) using:

```
$ make help
```